Table 5: Comparison of error detection performance (AUC) on Mistral-7B-Instruct.

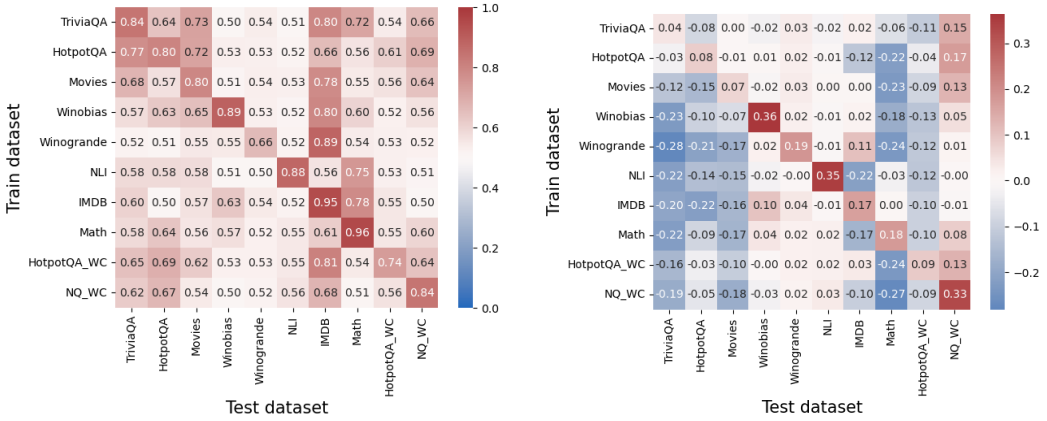
	Mistral-7B-Instruct				
	TriviaQA	Winobias	Math	Movies	IMDB
Logits-mean	0.60 ±0.009	0.56 ±0.017	0.55 ±0.029	0.63 ±0.005	0.57 ±0.006
Logits-mean-exact	0.68 ±0.007	0.54 ±0.012	0.51 ±0.005	0.70 ±0.004	0.87 ±0.007
Logits-min	0.63 ±0.008	0.59 ±0.012	0.51 ±0.017	0.66 ±0.008	0.52 ±0.007
Logits-min-exact	0.75 ±0.006	0.53 ±0.013	0.71 ±0.009	0.74 ±0.005	0.87 ±0.007
Logits-max	0.54 ±0.005	0.53 ±0.012	0.54 ±0.039	0.54 ±0.004	0.47 ±0.004
Logits-max-exact	0.55 ±0.004	0.54 ±0.011	0.32 ±0.015	0.61 ±0.006	0.87 ±0.007
Probas-mean	0.60 ±0.007	0.58 ±0.018	0.56 ±0.028	0.61 ±0.002	0.54 ±0.008
Probas-mean-exact	0.71 ±0.003	0.57 ±0.015	0.71 ±0.014	0.74 ±0.006	0.84 ±0.007
Probas-min	0.59 ±0.008	0.58 ±0.014	0.50 ±0.025	0.60 ±0.008	0.51 ±0.010
Probas-min-exact	0.74 ±0.004	0.57 ±0.016	0.75 ±0.011	0.73 ±0.006	0.84 ±0.007
Probas-max	0.50 ±0.006	0.41 ±0.010	0.53 ±0.009	0.51 ±0.005	0.48 ±0.004
Probas-max-exact	0.51 ±0.007	0.54 ±0.010	0.45 ±0.015	0.60 ±0.003	0.84 ±0.007
p(True)	0.68 ±0.005	0.45 ±0.021	0.48 ±0.026	0.62 ±0.005	0.62 ±0.009
p(True)-exact	0.74 ±0.003	0.40 ±0.021	0.60 ±0.025	0.69 ±0.008	0.60 ±0.009
Probe @ token					
Last generated [-1]	0.71 ±0.006	0.82 ±0.004	0.74 ±0.008	0.72 ±0.005	0.92 ±0.010
Before last generated [-2]	0.73 ±0.004	0.85 ±0.004	0.74 ±0.007	0.72 ±0.006	0.94 ±0.006
End of question	0.76 ±0.008	0.82 ±0.011	0.72 ±0.007	0.74 ±0.003	0.96 ±0.006
Exact answer last	0.85 ±0.004	**0.92** ±0.005	**0.92** ±0.008	0.81 ±0.003	**0.97** ±0.005
Exact answer last+1	**0.86** ±0.006	0.88 ±0.006	0.90 ±0.010	**0.82** ±0.003	0.96 ±0.006
	HotpotQA	HotpotQA-WC	Winogrande	NLI	NQ-WC
Logits-mean	0.61 ±0.002	0.55 ±0.009	0.59 ±0.004	0.64 ±0.006	0.71 ±0.008
Logits-mean-exact	0.66 ±0.009	0.55 ±0.004	0.49 ±0.004	0.57 ±0.004	0.69 ±0.009
Logits-min	0.61 ±0.003	0.53 ±0.013	0.61 ±0.003	0.62 ±0.002	0.67 ±0.008
Logits-min-exact	0.77 ±0.004	0.67 ±0.013	0.48 ±0.004	0.54 ±0.005	0.69 ±0.006
Logits-max	0.53 ±0.008	0.51 ±0.011	0.52 ±0.006	0.59 ±0.008	0.63 ±0.011
Logits-max-exact	0.51 ±0.011	0.41 ±0.010	0.49 ±0.007	0.64 ±0.003	0.63 ±0.013
Probas-mean	0.63 ±0.003	0.56 ±0.010	0.58 ±0.005	0.62 ±0.005	0.68 ±0.010
Probas-mean-exact	0.72 ±0.006	0.66 ±0.010	0.46 ±0.004	0.57 ±0.003	0.65 ±0.008
Probas-min	0.58 ±0.003	0.52 ±0.008	0.59 ±0.002	0.58 ±0.008	0.65 ±0.014
Probas-min-exact	0.76 ±0.004	0.68 ±0.010	0.46 ±0.005	0.57 ±0.003	0.66 ±0.008
Probas-max	0.50 ±0.005	0.53 ±0.003	0.48 ±0.007	0.52 ±0.007	0.51 ±0.005
Probas-max-exact	0.46 ±0.010	0.46 ±0.010	0.48 ±0.004	0.53 ±0.004	0.52 ±0.018
p(True)	0.54 ±0.006	0.54 ±0.004	0.53 ±0.003	0.58 ±0.003	0.57 ±0.006
p(True)-exact	0.60 ±0.008	0.48 ±0.005	0.57 ±0.011	0.65 ±0.004	0.57 ±0.009
Probe @ token					
Last generated [-1]	0.72 ±0.005	0.64 ±0.005	0.74 ±0.005	0.85 ±0.004	0.82 ±0.006
Before last generated [-2]	0.73 ±0.006	0.64 ±0.004	0.76 ±0.004	0.87 ±0.002	0.84 ±0.009
End of question	0.80 ±0.003	0.63 ±0.003	0.71 ±0.007	0.79 ±0.004	0.85 ±0.010
Exact answer last	0.85 ±0.003	0.75 ±0.006	**0.84** ±0.005	**0.93** ±0.003	0.86 ±0.003
Exact answer last+1	**0.85** ±0.002	**0.76** ±0.004	0.80 ±0.004	0.92 ±0.004	**0.87** ±0.006

Table 6: Comparison of error detection performance (AUC) on Llama-8b.

	Llama-8b				
	TriviaQA	Winobias	Math	Movies	IMDB
Logits-mean	0.58 ±0.006	0.44 ±0.015	0.43 ±0.026	0.64 ±0.008	0.77 ±0.007
Logits-mean-exact	0.63 ±0.007	0.50 ±0.015	0.50 ±0.028	0.64 ±0.008	0.77 ±0.007
Logits-min	0.75 ±0.007	0.50 ±0.022	0.45 ±0.042	0.73 ±0.005	0.73 ±0.007
Logits-min-exact	0.76 ±0.003	0.53 ±0.009	0.75 ±0.022	0.73 ±0.005	0.77 ±0.007
Logits-max	0.48 ±0.006	0.48 ±0.009	0.42 ±0.027	0.53 ±0.005	0.72 ±0.007
Logits-max-exact	0.52 ±0.007	0.49 ±0.014	0.35 ±0.026	0.53 ±0.005	0.77 ±0.007
Probas-mean	0.64 ±0.006	0.41 ±0.008	0.61 ±0.029	0.71 ±0.007	0.70 ±0.008
Probas-mean-exact	0.72 ±0.005	0.50 ±0.018	0.54 ±0.026	0.72 ±0.006	0.88 ±0.003
Probas-min	0.79 ±0.008	0.43 ±0.004	0.75 ±0.044	0.74 ±0.005	0.68 ±0.005
Probas-min-exact	0.82 ±0.003	0.53 ±0.014	0.78 ±0.022	0.74 ±0.005	0.88 ±0.003
Probas-max	0.49 ±0.006	0.50 ±0.009	0.46 ±0.032	0.53 ±0.007	0.60 ±0.009
Probas-max-exact	0.53 ±0.008	0.50 ±0.018	0.36 ±0.032	0.54 ±0.007	0.88 ±0.003
p(True)	0.62 ±0.005	0.48 ±0.011	0.53 ±0.027	0.61 ±0.005	0.51 ±0.010
p(True)-exact	0.67 ±0.002	0.53 ±0.017	0.63 ±0.028	0.58 ±0.005	0.52 ±0.008
Probe @ token					
Last generated [-1]	0.77 ±0.005	0.59 ±0.024	0.83 ±0.013	0.82 ±0.005	0.94 ±0.002
Before last generated [-2]	0.76 ±0.012	0.58 ±0.021	0.82 ±0.032	0.79 ±0.004	0.96 ±0.002
End of question	0.73 ±0.005	0.77 ±0.012	0.80 ±0.027	0.78 ±0.005	0.68 ±0.009
Exact answer last	**0.82** ±0.006	**0.91** ±0.007	**0.96** ±0.010	0.80 ±0.005	**0.97** ±0.001
Exact answer last+1	0.82 ±0.006	0.86 ±0.008	0.95 ±0.007	**0.82** ±0.006	0.95 ±0.003
	HotpotQA	HotpotQA-WC	Winogrande	NLI	NQ-WC
Logits-mean	0.65 ±0.004	0.62 ±0.006	0.48 ±0.003	0.47 ±0.002	0.53 ±0.010
Logits-mean-exact	0.55 ±0.003	0.54 ±0.006	0.49 ±0.004	0.48 ±0.002	0.58 ±0.009
Logits-min	0.57 ±0.004	0.49 ±0.003	0.48 ±0.003	0.48 ±0.007	0.58 ±0.009
Logits-min-exact	0.69 ±0.002	0.68 ±0.006	0.49 ±0.003	0.48 ±0.007	0.61 ±0.010
Logits-max	0.61 ±0.005	0.60 ±0.004	0.48 ±0.003	0.52 ±0.003	0.51 ±0.008
Logits-max-exact	0.47 ±0.003	0.46 ±0.005	0.49 ±0.004	0.51 ±0.002	0.54 ±0.005
Probas-mean	0.67 ±0.002	0.62 ±0.006	0.49 ±0.002	0.48 ±0.004	0.57 ±0.003
Probas-mean-exact	0.62 ±0.005	0.56 ±0.005	0.51 ±0.002	0.46 ±0.006	0.64 ±0.007
Probas-min	0.62 ±0.006	0.51 ±0.002	0.49 ±0.003	0.50 ±0.010	0.62 ±0.005
Probas-min-exact	0.76 ±0.005	0.67 ±0.004	0.51 ±0.002	0.50 ±0.010	0.69 ±0.008
Probas-max	0.61 ±0.004	0.58 ±0.004	0.48 ±0.002	0.48 ±0.003	0.51 ±0.012
Probas-max-exact	0.49 ±0.003	0.44 ±0.004	0.51 ±0.003	0.47 ±0.002	0.56 ±0.005
p(True)	0.52 ±0.007	0.45 ±0.005	0.54 ±0.004	0.54 ±0.007	0.56 ±0.006
p(True)-exact	0.58 ±0.005	0.50 ±0.007	0.64 ±0.004	0.62 ±0.005	0.61 ±0.002
Probe @ token					
Last generated [-1]	0.76 ±0.007	0.57 ±0.006	0.59 ±0.006	0.89 ±0.002	0.66 ±0.010
Before last generated [-2]	0.74 ±0.007	0.58 ±0.005	0.59 ±0.005	0.94 ±0.002	0.63 ±0.008
End of question	0.71 ±0.006	0.53 ±0.004	0.48 ±0.003	0.91 ±0.001	0.66 ±0.004
Exact answer last	0.81 ±0.006	0.77 ±0.004	**0.65** ±0.004	**0.94** ±0.002	**0.75** ±0.008
Exact answer last+1	**0.82** ±0.004	**0.79** ±0.001	0.57 ±0.004	0.90 ±0.002	0.75 ±0.007

Table 7: Comparison of error detection performance (AUC) on Llama-8b-Instruct.

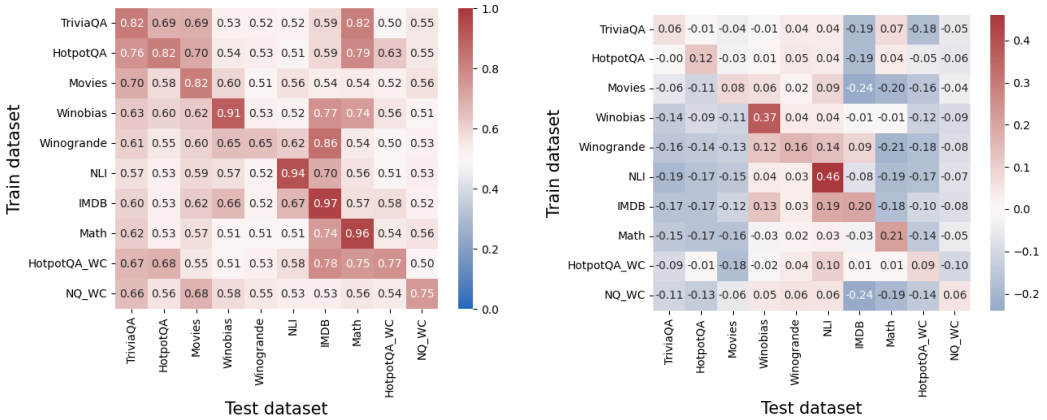
	Llama-8b-Instruct				
	TriviaQA	Winobias	Math	Movies	IMDB
Logits-mean	0.66 ±0.005	0.60 ±0.026	0.75 ±0.018	0.75 ±0.005	0.59 ±0.017
Logits-mean-exact	0.71 ±0.006	0.55 ±0.019	0.80 ±0.021	0.72 ±0.004	0.88 ±0.012
Logits-min	0.74 ±0.007	0.61 ±0.024	0.75 ±0.016	0.71 ±0.005	0.55 ±0.016
Logits-min-exact	0.79 ±0.006	0.61 ±0.019	0.89 ±0.018	0.77 ±0.006	0.88 ±0.012
Logits-max	0.54 ±0.007	0.55 ±0.013	0.73 ±0.027	0.67 ±0.003	0.51 ±0.009
Logits-max-exact	0.58 ±0.005	0.54 ±0.019	0.64 ±0.014	0.61 ±0.003	0.88 ±0.012
Probas-mean	0.67 ±0.006	0.63 ±0.024	0.66 ±0.033	0.73 ±0.006	0.73 ±0.015
Probas-mean-exact	0.75 ±0.009	0.61 ±0.014	0.83 ±0.022	0.74 ±0.005	0.74 ±0.021
Probas-min	0.67 ±0.009	0.65 ±0.019	0.64 ±0.036	0.65 ±0.004	0.57 ±0.016
Probas-min-exact	0.79 ±0.008	0.62 ±0.014	0.86 ±0.024	0.74 ±0.005	0.74 ±0.021
Probas-max	0.54 ±0.003	0.49 ±0.020	0.57 ±0.022	0.64 ±0.006	0.49 ±0.008
Probas-max-exact	0.56 ±0.007	0.55 ±0.016	0.57 ±0.018	0.61 ±0.003	0.74 ±0.021
p(True)	0.73 ±0.008	0.59 ±0.020	0.62 ±0.017	0.66 ±0.004	0.60 ±0.006
p(True)-exact	0.73 ±0.005	0.63 ±0.014	0.59 ±0.018	0.63 ±0.006	0.76 ±0.004
Probe @ token					
Last generated [-1]	0.81 ±0.005	0.86 ±0.007	0.82 ±0.016	0.78 ±0.004	0.81 ±0.014
Before last generated [-2]	0.75 ±0.005	0.88 ±0.005	0.79 ±0.020	0.82 ±0.005	0.83 ±0.006
End of question	0.77 ±0.007	0.80 ±0.018	0.72 ±0.023	0.76 ±0.005	0.87 ±0.006
Exact answer last	**0.83** ±0.002	**0.93** ±0.004	**0.95** ±0.027	0.85 ±0.005	**0.96** ±0.003
Exact answer last+1	0.83 ±0.006	0.90 ±0.005	0.94 ±0.023	**0.86** ±0.004	0.95 ±0.004
	HotpotQA	HotpotQA-WC	Winogrande	NLI	NQ-WC
Logits-mean	0.65 ±0.002	0.56 ±0.004	0.58 ±0.007	0.59 ±0.009	0.65 ±0.006
Logits-mean-exact	0.66 ±0.008	0.57 ±0.005	0.48 ±0.003	0.49 ±0.010	0.67 ±0.005
Logits-min	0.67 ±0.008	0.55 ±0.007	0.60 ±0.008	0.53 ±0.009	0.68 ±0.004
Logits-min-exact	0.76 ±0.010	0.65 ±0.010	0.48 ±0.004	0.50 ±0.009	0.68 ±0.004
Logits-max	0.59 ±0.005	0.56 ±0.005	0.46 ±0.004	0.55 ±0.013	0.56 ±0.006
Logits-max-exact	0.52 ±0.006	0.48 ±0.002	0.48 ±0.003	0.49 ±0.009	0.63 ±0.008
Probas-mean	0.61 ±0.002	0.56 ±0.010	0.57 ±0.007	0.58 ±0.007	0.65 ±0.007
Probas-mean-exact	0.68 ±0.008	0.65 ±0.006	0.51 ±0.006	0.57 ±0.009	0.67 ±0.003
Probas-min	0.60 ±0.004	0.51 ±0.007	0.59 ±0.007	0.55 ±0.005	0.64 ±0.008
Probas-min-exact	0.74 ±0.007	0.67 ±0.007	0.51 ±0.006	0.59 ±0.008	0.66 ±0.004
Probas-max	0.56 ±0.005	0.53 ±0.005	0.46 ±0.003	0.51 ±0.004	0.55 ±0.004
Probas-max-exact	0.49 ±0.007	0.47 ±0.002	0.51 ±0.005	0.50 ±0.009	0.62 ±0.006
p(True)	0.55 ±0.005	0.55 ±0.008	0.47 ±0.002	0.54 ±0.006	0.71 ±0.003
p(True)-exact	0.55 ±0.004	0.50 ±0.005	0.50 ±0.008	0.50 ±0.003	0.67 ±0.007
Probe @ token					
Last generated [-1]	0.77 ±0.005	0.68 ±0.006	0.69 ±0.006	0.78 ±0.005	0.77 ±0.009
Before last generated [-2]	0.76 ±0.002	0.69 ±0.005	0.67 ±0.008	0.79 ±0.004	0.75 ±0.007
End of question	0.78 ±0.004	0.60 ±0.003	0.65 ±0.004	0.74 ±0.002	0.75 ±0.011
Exact answer last	**0.83** ±0.005	**0.76** ±0.003	**0.78** ±0.007	**0.91** ±0.005	**0.78** ±0.006
Exact answer last+1	0.83 ±0.002	0.76 ±0.006	0.70 ±0.006	0.90 ±0.004	0.78 ±0.007



(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 7: Generalization between datasets, Mistral-7b.



(a) Raw AUC values. Values above 0.5 indicate some generalization.

(b) Performance (AUC) difference of the probe and the logit-based method. Values above 0 indicate generalization beyond the logit-based method.

Figure 8: Generalization between datasets, Llama-3-8b.

C Full Generalization Results

Figures 7, 8 and 9 present the generalization results for the remaining models. While these results exhibit similar high-level patterns to those found in the main paper on Mistral-7b-instruct, notable differences suggest that these models may possess different mechanisms for encoding truthfulness.

D Taxonomy of Errors

Figure 10 presents, for each amount of resamples, the amount percentage of answers for which at least one generated answer was correct. The experiment was done on Mistral-7b-instruct with the TriviaQA dataset. For many answers that the greedy decoding fails to correctly provide an answer, the LLM is still able to generate the correct answer in at least one resample. The plot plateues around 30 resamples.